

## Experiment 6 : Migrating a Web Application to Docker Containers

Migrating a Web Application to Docker Containers means **taking your existing app and packaging it into a container** so it runs the same way anywhere (local machine, server, cloud like AWS).

What is Docker?

Docker is a software tool that helps you:

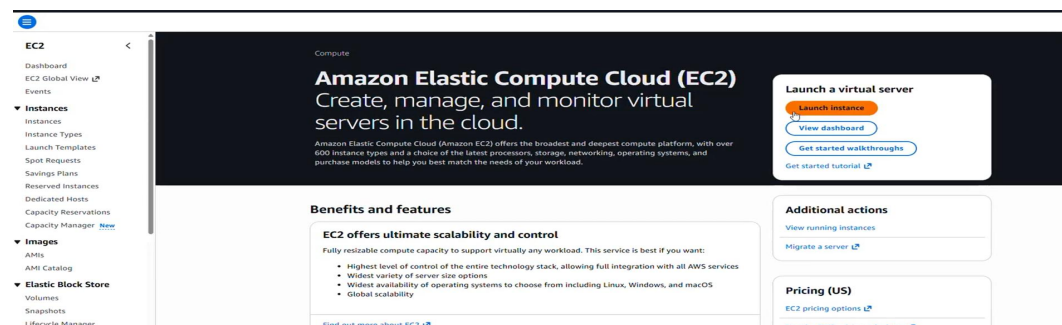
**Pack your application plus everything it needs into one box**  
That box is called a **container**

### Why Migrate to Docker?

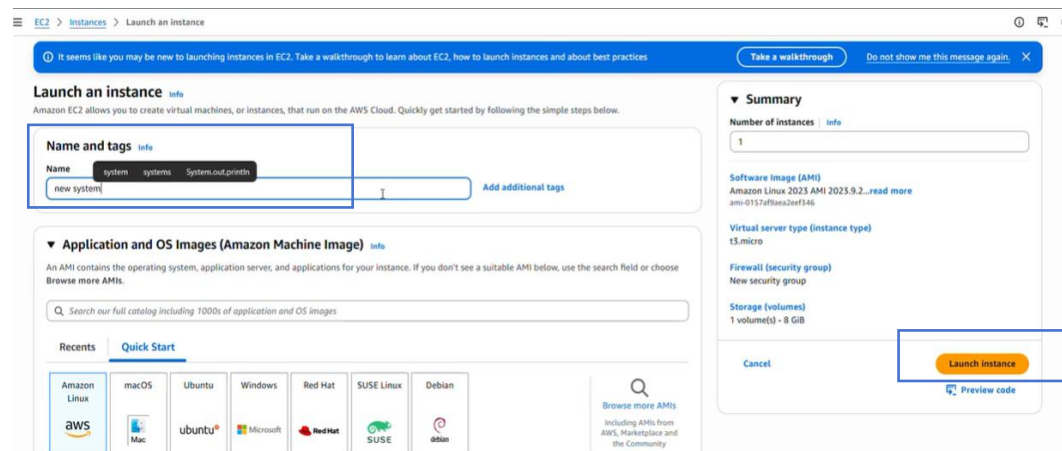
- Easy deployment
- Same environment everywhere
- Faster setup
- Scalable (works well with cloud)

### Step1 : Click Launch Instance

On your screen → click **Launch Instance**



### Step 2:- Give the Name and Tag for Instance and select key pair Login



EC2 > Instances > Launch an instance

64-bit (x86) uefi-preferred ami-0157af9aa2eef346 2025-11-04 ec2-user verified provider

**Instance type** [info](#) [Get advice](#)

Instance type: t3.micro  
 Family: t3 2 vCPU 1 GiB Memory Current generation: true  
 On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour  
 On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour  
 On-Demand Windows base pricing: 0.0196 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

**Key pair (login)** [info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required  
 Proceed without a key pair (Not recommended) Default value [Create new key pair](#)

**Network settings** [info](#) [Edit](#)

Network [info](#)

**Summary**

Number of instances: 1 [info](#)

Software Image (AMI)  
 Amazon Linux 2023 AMI 2023.9.2...[read more](#)  
 ami-0157af9aa2eef346

Virtual server type (instance type)  
 t3.micro

Firewall (security group)  
 New security group

Storage (volumes)  
 1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [View Pricing page](#)

#### Step 4: After Launch the instance we are connect to your Instance

RWS [Search] [Alt+S] United States (N. Virginia)

EC2 > Instances > Launch an instance

**Success**  
 Successfully initiated launch of instance i-098733d6309f8023

**Launch log**

**Next Steps**  
 What would you like to do next with this instance, for example "create alarm" or "create backup"

**Create billing usage alerts**  
 To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.  
[Create billing alerts](#)

**Connect to your instance**  
 Once your instance is running, log into it from your local computer.  
[Connect to instance](#)  
[Learn more](#)

**Connect an RDS database**  
 Configure the connection between an EC2 instance and a database to allow traffic flow between them.  
[Connect an RDS database](#)  
[Create a new RDS database](#)  
[Learn more](#)

**Create EBS snapshot policy**  
 Create a policy that automates the creation and deletion of EBS snapshots.  
[Create EBS snapshot policy](#)

**Connect** [info](#)

Connect to an instance using the browser-based client.

**EC2 Instance Connect** Session Manager SSH client EC2 serial console

Instance ID  
 i-098733d6309f8023 (new system)

**Connection type**

☒ Connect using a Public IP  
 Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP  
 Connect using a private IP address and a VPC endpoint

**Public IPv4 address**  
 54.162.157.198

**IPv6 address**  
 -

**Username**  
 Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.  
 ec2-user

**Note:** In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

[Cancel](#) [Connect](#)



#### Step 5:

# Step 1: Update system packages  
 sudo yum update -y

# Step 2: Install Docker

sudo yum install docker -y

# Step 3: Start the Docker service

sudo systemctl start docker

# Step 4: Enable Docker to start on boot

sudo systemctl enable docker

# Step 5: Pull and run the Nginx container

sudo docker run -d -p 80:80 nginx

```
ec2-user@ip-172-31-72-216 ~]$ sudo yum install docker -y
Last metadata expiration check: 0:01:42 ago on Mon Mar 30 16:09:12 2026.
Dependencies resolved.

```

| Package                  | Architecture | Version                | Repository  | Size  |
|--------------------------|--------------|------------------------|-------------|-------|
| Installing:<br>docker    | x86_64       | 25.0.14-1.amzn2023.0.2 | amazonlinux | 46 M  |
| Installing dependencies: |              |                        |             |       |
| container-selinux        | noarch       | 4:2.245.0-1.amzn2023   | amazonlinux | 58 k  |
| containerd               | x86_64       | 2.2.1-1.amzn2023.0.1   | amazonlinux | 24 M  |
| iptables-libs            | x86_64       | 1.8.8-3.amzn2023.0.2   | amazonlinux | 401 k |
| iptables-nft             | x86_64       | 1.8.8-3.amzn2023.0.2   | amazonlinux | 183 k |
| libgroup                 | x86_64       | 3.0-1.amzn2023.0.1     | amazonlinux | 75 k  |
| libnetfilter_conntrack   | x86_64       | 1.0.8-2.amzn2023.0.2   | amazonlinux | 58 k  |
| libnftnl                 | x86_64       | 1.0.1-19.amzn2023.0.2  | amazonlinux | 30 k  |
| libnftnl                 | x86_64       | 1.2.2-2.amzn2023.0.2   | amazonlinux | 84 k  |
| pigs                     | x86_64       | 2.5-1.amzn2023.0.3     | amazonlinux | 83 k  |
| runc                     | x86_64       | 1.3.4-1.amzn2023.0.2   | amazonlinux | 3.9 M |

```
Transaction Summary
Install 11 Packages
```

```
ec2-user@ip-172-31-72-216 ~]$ sudo systemctl start docker
ec2-user@ip-172-31-72-216 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service -> /usr/lib/systemd/system/docker.service.
ec2-user@ip-172-31-72-216 ~]$ sudo docker run -d -p 80:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
ec781dee3f47: Pull complete
bb3d0aa29654: Pull complete
b10ddf6557d6: Download complete
cde7a05ae428: Download complete
b87e3d84dbb5: Download complete
b189680c601f: Download complete
be815e07e569: Download complete
```

## Step 6 : Click On Instance ID

The screenshot shows the AWS Management Console interface. On the left, the 'Instances' menu is expanded. The main area displays a table of EC2 instances. One instance, named 'new system' with ID 'i-0987333d6309f8023', is highlighted. Below the table, the details for this instance are shown, including the Public IPv4 address '54.162.157.198'.

## Step 5:

- if your application is not opening in the browser, go to the AWS EC2 Dashboard
- Click on Security → Security Groups
- You will see a Security Group ID

## Click on the Security Group ID

- Go to Inbound Rules
- Click Edit Inbound Rules

EC2

Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role

Owner ID

Launch time

Security groups

▼ Inbound rules

Filter rules

Table with 7 columns: Name, Security group rule ID, Port range, Protocol, Source, Security groups, Desc

▼ Outbound rules

Filter rules

Table with 7 columns: Name, Security group rule ID, Port range, Protocol, Destination, Security groups, Desc

EC2 > Security Groups > sg-0060c029835b23665 - launch-wizard-2 > Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID

sg-0b63be204d672c85d

Type

SSH

Protocol

TCP

Port range

22

Source

Custom

0.0.0.0/0

Description - optional

Delete

Type

HTTP

Protocol

TCP

Port range

80

Source

Custom

0.0.0.0/0

Description - optional

Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to [nginx.org](#).  
To engage with the community please visit [community.nginx.org](#).  
For enterprise grade support, professional services, additional security features and capabilities please refer to [f5.com/nginx](#).

Thank you for using nginx.